Ritwij Aryan Parmar

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New York, NY; open to San Francisco onsite

Experience

- **Distributed Robotics and Networked Embedded Sensing (DRONES) Lab** — Buffalo, NY
  Research Aide – Robotics Systems — *Jul 2025 – Feb 2026*
  - Reduced mapping drift to under **1.2%** across **500 meters** of GPS-denied environments by building a **ROS2 visual SLAM** pipeline for Boston Dynamics Spot with Gaussian Splatting integration.
  - Improved **LiDAR and VIO fusion** for low-texture subterranean navigation, reducing trajectory estimation error by **30%** while sustaining **20 hertz** real-time state estimation.
  - Built a real-robot inspection workflow for culvert-assessment data, converting field videos into defect masks and reviewable polygon annotations across **1,400+** sampled frames with per-run IoU/F1 reports.
- **Tata Elxsi – Global automotive engineering firm** — Bengaluru, India
  Software Engineer Intern — *Jan 2024 – Jun 2024*
  - Migrated an autonomous vehicle software stack from **ROS1 to ROS2** and tuned **DDS QoS** behavior in CARLA, reducing inter-module latency by **40%** and achieving sub-100 millisecond communication latency.
  - Built an **Extended Kalman Filter** vehicle state estimator that improved positioning accuracy by **35%**, and implemented a safety-constrained route planner with deterministic state transitions and bounded-latency path generation.
  - Built an autonomy evaluation-data pipeline that converted simulator runs into detector-ready datasets, aligning RGB frames, depth maps, semantic labels, ego pose, and calibration metadata into validated 2D/3D annotation packages.
  - Automated failure mining across **120+** night, rain, occlusion, cut-in, and pedestrian-crossing simulation runs by flagging low-confidence detections, missed ground-truth matches, and label/calibration mismatches.
- **LLMate.ai – Seed-stage GenAI analytics startup** — Bengaluru, India
  Software Engineer Intern — *Apr 2023 – Jul 2023*
  - Built backend services with Spring Boot, RabbitMQ, Docker, and GitHub Actions for production data workflows, reducing p95 response latency by about **40%**.
  - Built a natural-language SQL backend over **50,000+** records with parser validation, read-only execution, row/time limits, query-cost controls, and result caching.

Projects

- **HelixServe – Low-Latency GPU Inference Server – GitHub** – Increased serving throughput **5.7x** from **175.8 to 1,007.8 tokens/sec** and reduced p95 latency **75%** from **3.28s to 0.80s** on GCP NVIDIA L4 by implementing paged KV cache, continuous batching, prefix caching, CUDA Graph decode, and runtime metrics.

Open-Source Engineering

- **Robotics/Perception and Data Systems – GitHub** – Contributed **30+ merged upstream PRs** across computer-vision evaluation, robotics-style telemetry/data tooling, vector search, observability databases, and AI infrastructure.
- **Roboflow Supervision, FiftyOne, Daft, Qdrant, GreptimeDB** – Fixed non-square OBB IoU in Roboflow Supervision and added configurable keypoint-evaluation behavior to FiftyOne; also improved Daft MCAP URL handling, Qdrant scalar-quantized L2 scoring, and GreptimeDB Prometheus metadata discovery.

Skills

Languages: Python, C++, Java, SQL
Robotics: ROS2, ROS1, SLAM, LiDAR/VIO Sensor Fusion, EKF, CARLA, DDS QoS, State Estimation, Path Planning, Robot Bring-up
Perception/Data: OpenCV, Camera/LiDAR Data Processing, 2D/3D Annotation Validation, Perception Metrics, Failure Mining, Sensor Calibration Metadata
Software: Linux, Git, Docker, FastAPI, Spring Boot, RabbitMQ, GCP, Cloud Run, CI/CD

Education

- **University at Buffalo – SUNY** — Buffalo, NY
  Master of Science in Computer Science and Engineering; GPA: 3.74/4.0 — *Jan 2025 – May 2026*
- **Manipal Institute of Technology** — Manipal, India
  Bachelor of Technology in Computer Science and Engineering — *Aug 2020 – Aug 2024*